9/10/25

** Genetic optimization algorithm.

1. Start.
2. Generate random ~~population~~ population of POP-size.
3. FOR each generation from 1 to MAX-GEN DO
4. Calculate the fitness of the each individual.
5. Slect the best parents.
6. Apply crossover to produce children.
7. Apply mutation the children.
8. Creat a new population using the children.
9. END FOR
10. Return the best solution.

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Feature Selection.

Dog, cat

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"Feature Selection"

To implement a Genetic algorithm for feature Selection to identify the most relevant features from an animal dataset and use them Classify an animal as either dog or cat.

Input: POP-size, MAX-GEN, animal dataset
Output: Best feature subset and predicted class

1. Start.
2. Select features such as ear shape, fur pattern, and body size.
3. Generate random population of feature subset
4. For generation = 1 to MAX-GEN DO
5. calculate the fitness of the each feature subset
6. Select parents based on fitness
7. Apply crossover to generate offspring
8. Apply mutation to the offspring
10. END FOR
11. Select the best feature subset
12. Use Selected features to classify the animal as Dog or cat.
13. Display the predicted class.
14. STOP.

Procedure
Output